



Program: BS Computer Science

Semester: Fall 2025

Credit Hours: 3+1

Pre-requisite(s): EE1005 Digital Logic Design

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Lab # 1: Introduction & Environment Setup

Objectives

- To understand the basics of assembly language and its applications.
- To get familiar with DOSBox, NASM, and AFD tools.
- To learn the environment setup for running assembly language programs.

Background Theory

What is Assembly Language?

Assembly language is a way to talk directly to the computer's brain (CPU).

- It uses **short words (called mnemonics)** like MOV, ADD, SUB instead of 1s and 0s.
- It is very close to machine language, so it runs **very fast**.

Why learn it?

- To understand how computers really work.
- To write programs for small devices (like microcontrollers).
- To make programs that are super fast.
- To learn how to debug and fix problems at the hardware level.

What is DOSBox?

Modern computers do not support old 8088 programs directly. So we need an alternative method, which is DOSBox. It is an emulator program that creates a DOS-like environment on modern operating systems (Windows/Linux/Mac).

- **DOSBox** is a program that creates a fake (virtual) old computer inside your modern one.
- It gives us a **DOS environment** where we can run old assembly programs easily.
- Think of it like a **time machine** that lets us run 1980s software today.

In our labs, we will use DOSBox to run and test our assembly programs.

What is NASM?

NASM stands for **Netwide Assembler**.

- It takes the assembly code we write (.ASM file) and converts it into machine code (a .COM or .EXE file).
- Without NASM, the computer cannot understand your program.
- Syntax is clean and simple for beginners.



Example:

- We write: MOV AX, 5 (in assembly).
- NASM changes it to 101100000000101 (binary code the CPU understands).

What is AFD?

AFD stands for **Advanced Full-Screen Debugger**. It is a debugger tool for DOS.

- It is a tool that lets you run your program step by step.
- You can watch what happens inside the CPU while your program runs:
 - Which instruction is running.
 - How registers (AX, BX, CX, DX, etc.) are changing.
 - How memory is being used.

Think of AFD as a **magnifying glass** that shows what your program is really doing inside the computer.

Installation & Setup Guide

Step 1 – Download Required Files

- 1) **Download NASM** (Netwide Assembler); this will convert your .asm program into a .com executable.
- 2) **Download AFD** (Advanced Full-Screen Debugger); this will allow you to run your program step by step and check registers.
- 3) **Download and Install DOSBox**; this will create a DOS environment to run your old-style assembly programs.

Step 2 – Create a Working Folder

- 1) Choose any drive (e.g., D:\)
- 2) Create a new folder (e.g., D:\Assembly)
- 3) Copy both **NASM** and **AFD** executable files into this folder

Your folder should look like this:

```
D:\Assembly
→ nasm.exe
→ afd.exe
```

Step 3 – Write Your Assembly Program

- 1) Open any text editor (**VS Code**, **Notepad++**, etc.)
- 2) Write your assembly program in it
- 3) Save the file in the **same folder** as program .asm

Your folder will now look like:

```
D:\Assembly
→ nasm.exe
→ afd.exe
→ program.asm
```



Step 4 – Open DOSBox

- 1) Start **DOSBox** from your PC
- 2) Mount your folder:

```
mount c d:\assembly
d:
dir
```

You should see nasm.exe, afd.exe, and your program.asm.

Step 5 – Assemble and Run Program

- 1) Assemble your program using NASM: `nasm program.asm -o program.com`
This creates *program.com* (the executable file).
- 2) Run your program: `program.com`

Step 6 – Debug Program

- 1) To check registers and run step by step: `afd program.com`
- 2) Inside AFD, you can see instructions and values in registers like AX, BX, etc.

Tasks

- a) Explore DOSBox and mount the folder
- b) Create a file of assembly language `.asm`
- c) Run the Sample Program
- d) Explore AFD